AmbuRouteAI

🚑 An AI-Powered Traffic Management for Emergency Vehicles.

🚦 AmbuRouteAI is an AI-powered traffic management system that detects ambulances in real-time and dynamically controls traffic signals to provide a clear path for emergency vehicles. Using YOLOv8 for object detection and OpenCV for traffic light simulation, this system helps reduce ambulance response time and enhances emergency healthcare logistics.

🔧 Features

✅ Real-time Ambulance Detection – Uses YOLOv8 AI to identify ambulances from live traffic feeds. ✅ Smart Traffic Signal Control – Dynamically changes traffic lights using OpenCV when an ambulance is detected. ✅ Live Video Processing – Works with webcams or traffic camera feeds to analyze road conditions. ✅ Seamless Integration – Can be extended to IoT-enabled smart city infrastructure. ✅ Scalable & Efficient – Future-ready for integration with Google Maps API for real-time route optimization.

🛠 Tech Stack

Component Technology/Tool AI Model YOLOv8 (Ultralytics) Computer Vision OpenCV, NumPy Backend Python Traffic Simulation OpenCV Live Video Input Webcam / CCTV Feed

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📜 How It Works

1. AI-Powered Detection – YOLOv8 identifies ambulances in real-time from traffic video feeds. 2. Traffic Light Control – If an ambulance is detected, the traffic light turns green; otherwise, it remains red. 3. Visual Alerts & UI – The system draws bounding boxes around detected ambulances and simulates traffic signals. 4. Extensibility – The project can integrate with Google Maps API & IoT sensors for smarter city-wide traffic control.

📂 Project Structure

📦 AmbuRouteAI |── 📜 README.md # Project documentation |── 📜 requirements.txt # Required dependencies |── 📜 main.py # Main Python script |── 📦 models # YOLO model files |── 📂 media # Sample images/videos |── 📂 datasets # Training datasets (optional)

📥 Installation & Setup

1️⃣ Clone the Repository

git clone https://github.com/mantreshkhurana/AmbuRouteAI.git cd AmbuRouteAI

2️⃣ Install Dependencies

pip install -r requirements.txt

3️⃣ Download YOLOv8 Model

Download the pre-trained YOLOv8 model from Ultralytics and place it in the models folder.

wget https://github.com/ultralytics/assets/releases/download/v0.0.0/yolov8n.pt

4️⃣ Run the Project

To test with a sample traffic video:

python main.py

To run with live webcam:

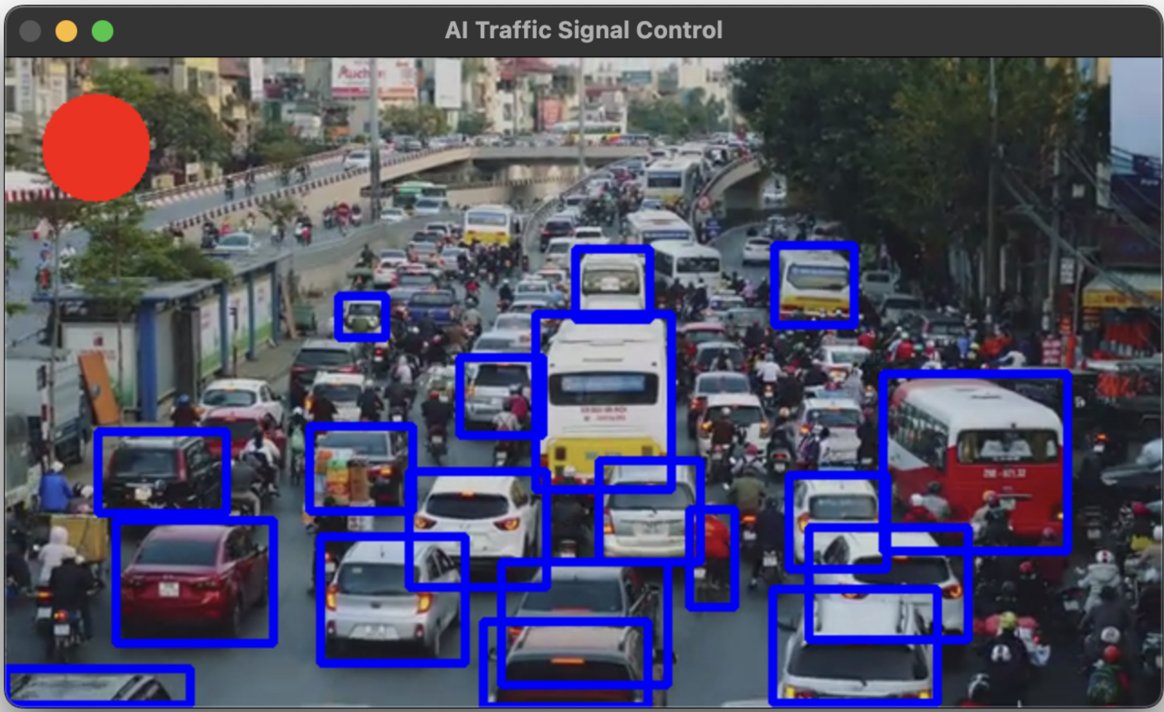
python main.py --source 0

🎯 Future Enhancements

- [ ] Integrate with Google Maps API – Real-time traffic data for route optimization.
- [ ] IoT-Based Smart Traffic Signals – Connect with Raspberry Pi & Arduino for real-world applications.
- [ ] Hospital Alert System – Send emergency notifications to hospitals about incoming patients.
- [ ] Mobile App Interface – Develop an ambulance tracking app with live traffic updates.

📸 Demo & Screenshots

📌 Detecting an ambulance and switching the traffic light in real-time.



Authors

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